

Kritarth Dandapat

Incoming PhD Student, Michigan State University | Undergraduate Researcher, University at Buffalo

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Professional Summary

I am a computer science researcher who genuinely enjoys building things that matter — whether that is a mobile app that helps elderly patients get oral health screenings, a system that tracks orthodontic progress remotely, or models that help scientists discover new materials faster.

I am completing an accelerated three-year BS in Computer Science (Minor: Statistics) at the University at Buffalo with a 3.8+ GPA, and I am starting my PhD at Michigan State University in Fall 2026, joining the Data Mining Laboratory under [Dr. Pang-Ning Tan](#). My doctoral research will focus on spatiotemporal machine learning, deep learning-based weather forecasting, and AI adversarial robustness.

What sets me apart is that I sit at the intersection of research and engineering. I have co-authored papers in digital health and computational materials science, while simultaneously shipping production mobile apps, deploying cloud infrastructure on AWS, and implementing advanced ML models like equivariant graph neural networks. I move quickly from idea to working system.

I have been recognized with the PEARL undergraduate research award (\$2,500) and placed second at the UB Health Futures Challenge (\$1,000).

Education

PhD in Computer Science

Michigan State University

Starting Fall 2026

East Lansing, MI

- Department of Computer Science and Engineering; Data Mining Laboratory (Advisor: [Dr. Pang-Ning Tan](#))
- Planned research focus: spatiotemporal machine learning, deep learning-based weather forecasting (DLWF), and AI adversarial robustness

Bachelor of Science in Computer Science, Minor in Statistics

University at Buffalo, SUNY

August 2023 – June 2026

Buffalo, NY

- Specialization in artificial intelligence; coursework spanning reinforcement learning, computer vision, machine learning, quantum computing, and distributed systems
- GPA 3.8+; Dean's List all semesters; accelerated three-year completion (21–22 credit semesters)
- Presidential Scholarship: \$15,000 per year; PEARL Award: \$2,500 competitive undergraduate research grant (Experiential Learning Network)

Research Experience

Research Assistant

Embedded Sensing and Computing (ESC) Group, University at Buffalo

Mobile sensing for healthcare and rehabilitation with [Prof. Wen Yao Xu](#).

June 2024 – Present

Buffalo, NY

OralScan: AI-Assisted Geriatric Oral Screening

June 2024 – February 2025

- Architected a full-stack mobile health platform (React Native, React web) integrating end-to-end **YOLOv8** vision pipelines for guided intraoral image acquisition, multi-class dental disease screening, and automated tooth numbering across temporal scan sessions.
- Formulated a spatio-temporal scan-guidance system that aggregates sequential intraoral frames to produce consistent disease-detection signals robust to user-induced motion variability.
- Co-authored formative usability and acceptability study submitted to *Smart Health* (under review); delivered oral presentations to clinical stakeholders at CTSI and SURC 2024.
- Awarded second place (\$1,000), University at Buffalo Health Futures Challenge (Spring 2026).

OrthoScan: Orthodontic Remote Monitoring

February 2025 – October 2025

- Engineered mobile and backend systems for at-home orthodontic progress tracking, deploying **YOLO**-based hardware detection paired with depth-assisted spatial measurement pipelines to quantify structural change across longitudinal patient visits.
- Reconstructed and analyzed **3D dental structures** from intraoral scans using pre-trained **GNNs** for 3D mesh processing; computed quantitative tooth position changes in both 2D image space and 3D world coordinates to track orthodontic progress with geometric precision.
- Translated sequential intraoral imagery into calibrated geometric progress signals, enabling quantitative remote monitoring between in-office appointments and reducing reliance on subjective clinician assessment.

mRehab: Telerehabilitation Platform ([mrehab.agency](#))

November 2025 – Present

- Provisioned and configured **AWS** backend infrastructure to host testing and staging environments, enabling AI agent capabilities to operate within the live application stack.
- Deployed a real-time **live sensor logging site** for continuous ingest and monitoring of wearable sensor streams during active therapy sessions.
- Implemented **Kalman filters** for sensor fusion and activity detection, improving signal smoothness and robustness of exercise recognition under noisy real-world IMU conditions.
- Contributed to session logging, multi-modal sensor ingest pipelines, and exercise scoring algorithms deployed in live outpatient therapy workflows.

Undergraduate Researcher

Peng Research Lab, University at Buffalo

Computational materials science and clean-energy applications with [Prof. Jiayu Peng](#).

June 2025 – Present

Buffalo, NY

Perovskite Ordering and Symmetry-Aware Graph Neural Networks

June 2025 – November 2025

- Benchmarked symmetry-aware **GNN** architectures (including **ALIGNN**) on ordering-dependent energetics in crystalline perovskites; contributed benchmarking results and revised analysis to arXiv:2409.13851 and the open-source **PerovskiteOrderingGCNNs** repository.
- Migrated experiment tracking infrastructure from SigOpt to **Weights & Biases**; automated training loops, hyperparameter sweeps, and multi-architecture evaluation pipelines, reducing manual overhead and improving reproducibility.
- Co-authored commentary on agentic AI for multimetallic catalyst discovery, published on *ChemRxiv* (2025).

Equivariant MLIPs, Alchemical Extensions, and Universal Atom Models

December 2025 – Present

- Implement **E(3)-equivariant** message-passing machine learning interatomic potentials (**MLIPs**) augmented with alchemical graph extensions to enable efficient composition sweeps and energy prediction over disordered crystalline structures.
- Integrate alchemical treatment with universal atom model (UMA) stacks (**MACE**) to accelerate structural relaxation and disorder characterization workflows in computational materials discovery pipelines.

Founder, Inference Foundry

Open Research Software Initiative

Starting May 2026

Remote

- Launched a collaborative open-source initiative building reproducible ML inference tooling, standardized research software scaffolding, and shared contributor infrastructure for the scientific ML community ([inference-foundry.rweb.site](#)).

Publications and Preprints

- Soni, P., **Dandapat, K.**, Gherardi, A., Bo, W., Li, R., and Xu, W. (2025). OralScan, an AI-Powered Mobile Tool for Geriatric Oral Healthcare: A Formative Usability and Acceptability Study. Submitted to *Smart Health* (under review).
- Peng, J., Liu, C., Luo, Y., and **Dandapat, K.** (2025). Accelerating Multimetallic Catalyst Discovery with Robotics and Agentic AI. *ChemRxiv*, version 1. DOI: [10.26434/chemrxiv-2025-13n3f](https://doi.org/10.26434/chemrxiv-2025-13n3f).

- Peng, J., et al. (2024). Learning Ordering in Crystalline Materials with Symmetry-Aware Graph Neural Networks. *arXiv* preprint [arXiv:2409.13851](https://arxiv.org/abs/2409.13851). (Contributed benchmarking and revisions.)

Research Presentations

- Oral presentation and live demo – CTSI Research Group of Doctors, Nurses, and Medical Students (April 2024)
- Oral presentation – SUNY Undergraduate Research Conference (SURC) (April 2024)

Selected Projects

PaddockPsychRL: F1-Informed Multi-Agent Reinforcement Learning [[GitHub](#)] *Spatiotemporal Modeling & Reinforcement Learning*

- Formulated a multi-agent RL framework integrating Formula 1 temporal performance signals into agent psychology models within **PettingZoo** cooperative environments, mapping real-world behavioral dynamics to learned policy structure.
- Constructed driver-specific agent profiles from **FastF1** 2025 lap-time consistency metrics, encoding temporal performance variability as per-agent action-noise scales to reflect authentic decision-making under pressure.
- Benchmarked tabular Q-learning and **MAPPO** baselines against psych-conditioned policies; psych-aware agents improved Strategy Resilience Score from 0.54 to 0.63 (+16.7%) under noisy multi-agent coordination.

Stack: Python, FastF1, PettingZoo, Ray RLlib, MAPPO, Q-learning

Marine Guardian: Ship Detection in Satellite Imagery *Geospatial Computer Vision & Deep Learning*

- Designed a maritime monitoring pipeline grounded in geometric computer vision principles before applying deep models, achieving low-latency inference on high-resolution satellite imagery for spatial object detection.
- Engineered a roundness-based geometric classifier paired with **MobileNetV2**, attaining 98.72% ship-vs.-non-ship classification accuracy while eliminating heavy encoder overhead unsuitable for edge deployment.
- Benchmarked **EfficientNet** transfer learning with a custom decoder against ResNet-50 feature extraction and K-means clustering, systematically validating architecture decisions across spatial generalization scenarios.

Stack: Python, OpenCV, TensorFlow, EfficientNet, MobileNetV2, ResNet-50, K-means

People Counting via CSRNet Density Estimation *Deep Learning & Spatial Regression*

- Developed a density-estimation pipeline for crowd counting in highly occluded surveillance scenes where bounding-box detectors fail due to extreme pedestrian overlap, framing the task as spatial regression over density maps.
- Trained **CSRNet** end-to-end in **PyTorch** to predict per-pixel crowd density fields, enabling robust aggregate counts from integrated spatial density rather than fragile per-instance localization.
- Accelerated training and inference with **CUDA**; preprocessed surveillance-style frames with **OpenCV** for repeatable benchmarking across public crowd-counting datasets.

Stack: PyTorch, CSRNet, CUDA, OpenCV

Human Emotion Detection: Multi-Architecture Benchmark *Computer Vision & Model Robustness*

- Constructed a controlled multi-architecture benchmark comparing shallow CNN, **ResNet-34**, and **Vision Transformer (ViT)** backbones on facial expression recognition under a unified preprocessing and evaluation protocol, isolating architectural contribution from data effects.
- Optimized training schedules and augmentation strategies per architecture; ViT and ResNet-34 variants demonstrated superior generalization over the CNN baseline on held-out facial expression data.
- Achieved 87.5% top-1 classification accuracy on the best-performing configuration after systematic cross-architecture ablation.

Stack: PyTorch, TensorFlow, CNN, ResNet-34, Vision Transformer, OpenCV

Professional Experience

Teaching Assistant, CSE 341: Computer Architecture

Department of Computer Science and Engineering, University at Buffalo

January 2026 – May 2026

Buffalo, NY

- Conducted weekly tutorial sessions on **Assembly** and **VHDL** for 30+ students, live-coding solutions to architecture problems including cache design, memory hierarchies, and instruction pipelining.
- Led recitations walking through processor pipeline stages, hazard detection, and cache replacement policies; developed worked examples and problem sets to reinforce lecture material.
- Held weekly office hours; graded exams and assignments; supported 100+ students asynchronously on Piazza throughout the semester.

Tutor and Peer-Assisted Leader (PAL)

Tutoring and Academic Support Services (TASS), University at Buffalo

August 2024 – December 2025

Buffalo, NY

- Served as **PAL for STA 119** (Introduction to Statistics): designed weekly worksheets aligned to recent lectures, then led two 50-minute collaborative sessions per week solving and explaining problems with students; course feedback indicated approximately 30% improvement in quiz performance.
- Provided one-to-one 1-hour tutoring appointments across seven courses: **STA 119, CSE 115, CSE 116, CSE 220, MTH 141, MTH 142**, covering introductory programming, data structures, and calculus.

Founder and Vice President

UB National Student Data Corps (NSDC)

October 2023 – May 2024

Buffalo, NY

- Co-founded the UB chapter; launched the chapter website, organized data science workshops, and built networking programs connecting undergraduates to research and industry opportunities.

Honors and Awards

- **PEARL Award** – \$2,500 competitive undergraduate research grant, University at Buffalo Experiential Learning Network (November 2025)
- **Health Futures Challenge** – Second place (\$1,000) for OralScan, University at Buffalo (Spring 2026)
- **Cybersecurity Excellence** – Top 100 globally, Northeastern Cybersecurity C2C Finals; 10th in world finals (2025)
- **Collegiate Lockdown** – Top two teams representing UB; 4th in finals (2025)
- **Dean's List** – University at Buffalo (Fall 2023, Spring 2024, Fall 2024)
- **Presidential Scholarship** – \$15,000 per year

Competitions and Challenges

- IAA AppXcelerate Application, University at Buffalo (January–March 2025) – OralScan
- Russell L. Agrusa CSE Student Innovation Competition (November 2024) – team member, OralScan
- Aging Innovations Challenge (November 2024) – team member, OralScan
- Community Champions for Disability Health Challenge (October 2024) – OralScan

Technical Skills

- **Languages:** Python, JavaScript, Java, C++, Rust
- **Machine learning and AI:** PyTorch, TensorFlow, computer vision, deep learning, graph neural networks (GNNs), YOLOv8, Vision Transformer (ViT), reinforcement learning (MAPPO, Q-learning), Ray RLlib
- **Scientific ML and materials:** ALIGNN, machine learning interatomic potentials (MLIPs), MACE, E(3)-equivariant MPNNs, Weights & Biases, PyTorch Geometric
- **Software engineering:** React, React Native, Node.js, Django, Git, CUDA, OpenCV, NumPy, Matplotlib
- **Databases and cloud:** MongoDB, SQL, Firebase, SQLite, AWS, Google Colab, Firebase Hosting

Research Interests

- Spatiotemporal machine learning and deep learning-based weather forecasting (DLWF)
- AI adversarial robustness and out-of-distribution generalization
- Geometric and equivariant deep learning; graph neural networks for structured data
- Computer vision and sequence modeling for healthcare and scientific applications
- Machine learning for materials science and scientific discovery

Certifications

- Machine Learning Specialization – Stanford University and DeepLearning.AI (Coursera)
- Deep Learning Specialization – Stanford University and DeepLearning.AI (Coursera)
- PyTorch for Deep Learning – Udemy
- Additional professional coursework: deep learning (TensorFlow), Django, React, Node.js, Python (Udemy)